

# Prestige vs. Price: Spatial Capitalization of Higher Education at State Borders

Evidence from a Difference-in-Discontinuities Design (2012–2024)

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April 9, 2026

# Higher Education at the State Border

- State borders create sharp, discontinuous "cliffs" in university access.
- Crossing a state line changes your access to:
  - ① **Financial Subsidies:** Tens of thousands of dollars in in-state tuition discounts.
  - ② **Prestige:** Potential access to elite, globally-ranked public flagship institutions.
- **The Question:** Do households spatially sort to capture these benefits, and is this access capitalized into local housing markets?

# The Two Competing Hypotheses

## Hypothesis 1: Financial Cost Driven

Housing markets price the pure financial savings of the college "tuition cliff." Families sort to capture cheaper in-state pricing.

## Hypothesis 2: Prestige Option Value

Housing markets price the *option value* of access to elite public flagship tiers (e.g., Top 50 institutions), regardless of the exact financial cost.

# Theoretical Framework: Spatial Supply and Demand

- **Setup:** Two states (A and B) separated by a border  $x = 0$ .
- **Mechanism:** Housing supply  $S(P)$  is fixed in the short run at exactly the border line.
- Residential demand  $D(P, A_s)$  depends on housing price  $P$  and state-specific amenities  $A_s = f(\text{Tuition}_s, \text{Caliber}_s)$ .
- **Intuition:** If State A randomly gains a Top 50 university, its amenity value  $A_A$  spikes.
- Demand shifts across the border from State B into State A. Since supply at the border is inelastic, land prices in A must rise discontinuously relative to B to clear the market.

# Preview of Main Results

- 1 **The "Tuition Illusion":** The financial cost of in-state tuition is **not** capitalized into housing once local property and income tax regimes are absorbed.
- 2 **The "Prestige Premium":** Access to a Top 50 public university yields a precisely estimated **9.8% housing premium** immediately across the state line.
- 3 **Non-Linear Option Value:** Capitalization follows an inverse U-shape. The market ignores generic Top 100 schools (too broad), but violently values elite Top 50 option value.

# The Identification Challenge

- **Naive OLS:** Biased by massive regional unobservables (e.g., Silicon Valley vs. Nevada deserts).
- **Spatial Regression Discontinuity (RD):** Assumes unobservables change smoothly at the border. But states *simultaneously* change property taxes, income taxes, and school quality exactly at the border.
- We cannot isolate the higher education effect using cross-sectional RD alone.

# The Spatial Engineering: Segmenting the Border

- **The Naive OLS Problem:** Standard OLS compares a house in Silicon Valley to a house in the Nevada desert.
- **The Border Segment Solution:** We fracture the entire US state-border network into discrete, evenly distributed segments ( $\approx 50\text{km}$  bands).
- We restrict comparisons *strictly* to housing units that share the exact same microscopic border segment, effectively equalizing unobserved spatial characteristics like weather, regional highway access, and geographic macro shocks.

# Methodology: Bandwidth and MSE-Optimality

- **How far from the border is valid?**
- We compute data-driven, MSE-optimal geographic bandwidths ( $h^*$ ) using Triangular Kernels (via `rdrobust`).
- The optimization algorithm iterates segment-by-segment calculation, falling back to dyadic-level optimization if specific border segments are too rural (sparse density).
- This rigorously filters the 2.7M initial dataset down to the 448k highly resilient geospatial estimation pool.



# The Spatial DiDC Architecture

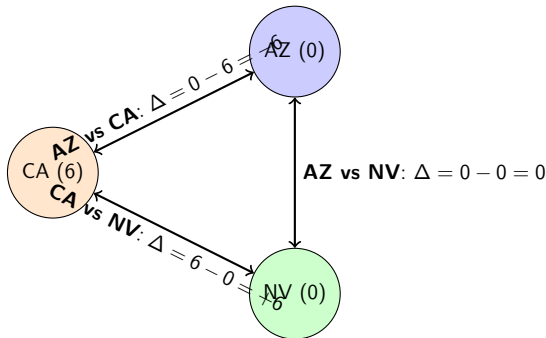
## Econometric Specification

$$\ln(P_{zt}) = \beta(S_i \times \Delta\text{Policy}_{st}) + \gamma X_{zt} + \alpha_{sy} + \epsilon_{zt}$$

- $S_i$ : Spatial dummy (1 if Treated Side, 0 if Control Side).
- $\Delta\text{Policy}_{st}$ : The dyadic gap in Tuition or Caliber.
- $\alpha_{sy}$ : **Border-Segment-by-Year Fixed Effects**. Absorbs all cross-border temporal and unobserved spatial shocks specific to that exact junction.
- $\gamma X_{zt}$ : Time-varying controls (TAXSIM liabilities, Property Taxes, local GDP).
- **Two-Way Clustered Standard Errors** (ZCTA and Border Dyad).

# Visualizing Alphabetical Symmetry (CA vs AZ vs NV)

**The Assignment Problem:** For any state dyad, which state is  $S_i = 1$ ? We arbitrarily sort alphabetically. The math behaves symmetrically.



- **Proof of Validity:** If CA is control ( $S_{AZ} = 1$ ), the parameter  $\Delta$  is negative ( $-6$ ). If CA is treated ( $S_{CA} = 1$ ),  $\Delta$  is positive ( $+6$ ).
- The  $S_i$  dummy interaction term perfectly absorbs the polarity, identifying the absolute spatial premium strictly derived from the amenity gap magnitude.

- **Tiebout Sorting (1956) & School Capitalization:** Black (1999) proves housing markets capitalize local K-12 school quality at attendance boundaries.
- **The Gap:** We know little about whether families sort at the *macro-state level* for higher education.
- **Value Added:**
  - ▶ Extends Tiebout sorting from K-12 to the higher-education market.
  - ▶ Overcomes policy endogeneity typical of macro-state comparisons by using a high-resolution Spatial Difference-in-Discontinuities (DiDC) estimator across 448k estimation observations.

# Data Curation: Housing & Education Policy

## 1. Housing Outcomes (2012–2024)

- ZCTA-level panel from Redfin (Median Price per Square Foot).
- Derived distances to the boundary ( $d_{ic}$ ) for precise spatial sorting.

## 2. Higher Education Policy

- **Financial:** IPEDS panel for weighted state tuition averages.
- **Prestige:** U.S. News & World Report historical rankings, algorithmically binned into Top 20, Top 50, and Top 100 state flagship caliber indicators.

# Data Curation: NBER TAXSIM (The Tiebout Controls)

- **The Problem:** High-amenity states charge high taxes. Failing to control for tax shifts across the border poisons the tuition coefficient.
- **The Solution (NBER TAXSIM):**
  - ▶ We simulated a standardized "Median Household" (\$100k income, married, 2 dependents).
  - ▶ Fed this identical family through the tax code of all 50 states natively for every year.
  - ▶ Isolated *pure statutory tax liability variation* without endogeneity from actual local wage or employment differences.

# Summary Statistics (Estimation DiDC Panel)

Table: ZCTA-Level Estimation Sample Properties (Filtered to Bandwidth)

| Variable                            | Observations | Mean    | Std. Dev. |
|-------------------------------------|--------------|---------|-----------|
| Median Price Per SqFt (\$)          | 268,224      | 163.22  | 367.17    |
| Distance to Border $d_{ic}$ (km)    | 448,941      | 0.43    | 36.82     |
| $\Delta$ In-State Tuition Gap (\$)  | 448,941      | -163.74 | 3390.95   |
| $\Delta$ Top 50 Caliber Gap (Count) | 344,411      | -0.11   | 0.88      |
| Effective Property Tax Rate         | 407,324      | 0.012   | 0.007     |
| Unemployment Rate (%)               | 444,795      | 5.071   | 2.174     |
| Real GDP Per Capita (\$)            | 444,795      | 58,720  | 1,050,974 |
| TAXSIM Federal Income Tax (\$)      | 419,176      | 198.88  | 6754.56   |
| TAXSIM State Income Tax (\$)        | 419,176      | 1959.85 | 1924.20   |

- Taxes simulated for a \$100k median household; GDP and Unemployment captured at ZCTA-level.

# Result I: The "Tuition Illusion"

Table: Capitalization of Tuition (Dependent Var: log PPSF)

|                             | (1) Spatial RD     | (2) Full DiDC            |
|-----------------------------|--------------------|--------------------------|
| Tuition Gap LATE (\$1k)     | -0.043*<br>(0.022) | -0.008<br>(0.013)        |
| Effective Property Tax Rate |                    | -19.494***<br>(3.081)    |
| State/Fed Income Taxes      |                    | Included                 |
| Fixed Effects               | None               | Segment_ID $\times$ Year |
| Observations                | 196,632            | 128,420                  |
| $R^2$ Adj.                  | 0.039              | 0.762                    |

- **Finding:** Raw Spatial RD assumes cheaper tuition drives higher prices (Col 1). Saturated DiDC models prove this is an illusion driven entirely by omitted state tax structures (Col 2).

## Result II: The Joint Capitalization Model

Table: Joint Model: Tuition vs. Top 50 Prestige (Dep Var: log PPSF)

|                           | (1) Base DiDC        | (2) Full DiDC (With Taxes) |
|---------------------------|----------------------|----------------------------|
| Caliber Gap LATE (Top 50) | 0.093**<br>(0.041)   | <b>0.098***</b><br>(0.031) |
| Tuition Gap LATE (\$1k)   | -0.026***<br>(0.009) | -0.008<br>(0.013)          |
| Fixed Effects             | Seg $\times$ Year    | Seg $\times$ Year          |
| Standard Errors           | Two-Way C.           | Two-Way C.                 |
| Observations              | 135,402              | 128,420                    |

- **Concrete Magnitude:** Gaining in-state access to a Top 50 university increases local housing prices by a precisely estimated **9.8%** ( $p < 0.01$ ) exactly at the state border.



# Exploring Option-Value Non-Linearities

- Does the prestige tier matter? Yes. We observe a structural inverse U-shape in the willingness to pay for elite access.

Table: LATE Premium by Flagship Caliber Tier (Full DiDC / Two-Way SE)

| Caliber Tier                  | Coefficient     | Std. Error |
|-------------------------------|-----------------|------------|
| Top 100 Gap (Generic Publics) | 0.028           | (0.027)    |
| Top 50 Gap (Elite Flagships)  | <b>0.098***</b> | (0.031)    |
| Top 20 Gap (Hyper-Elite)      | 0.010           | (0.048)    |

- Intuition:** Top 100 is too broad to induce cross-state sorting. Top 20 is mathematically too sparse geographically to generate signal. Top 50 bounds the precise threshold of the market's psychological valuation.

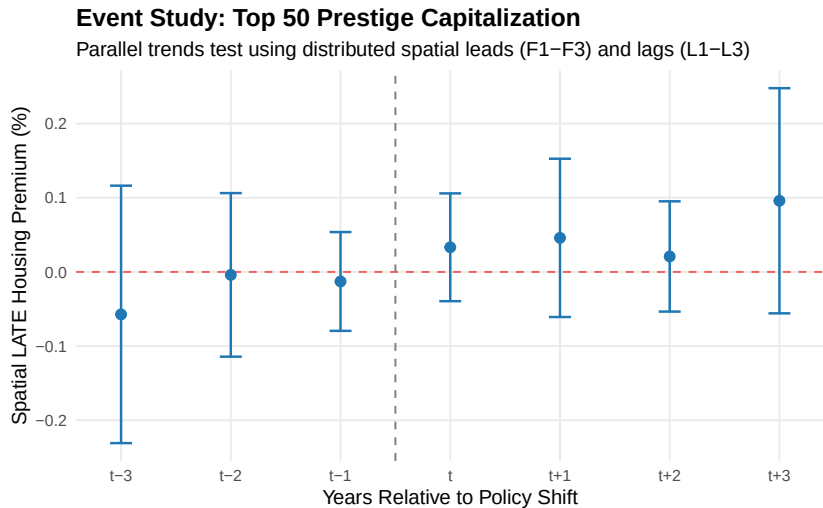
# Caveats & Limitations

- **Local Average Treatment Effect (LATE):** The 9.8% premium is intensely localized to the immediate geospatial state border. Housing markets deep in the interior of a state possess considerably lower spatial elasticity due to commuting limits.
- **Prestige Definition:** U.S. News Top 50 is a known, highly publicized public proxy. The genuine structural option value might operate on specific major availability (e.g., highly restricted elite Engineering or Pre-Med programs) embedded solely within those Top 50 schools.

# Conclusion & Policy Implications

- **Summary Results:** Spatial equilibrium across US borders heavily prices higher-education prestige, completely neutralizing standard in-state tuition financial savings.
- **Implication for Policymakers:**
  - ▶ Broadly subsidizing generic higher education does *not* attract cross-border migration or inflate the local tax base.
  - ▶ True geographic brain-drain prevention and profound housing capitalization require sustained, concentrated investment into an elite, globally-recognized flagship capable of breaching the Top 50 threshold.

# Appendix: Event Study Validation Plan



# Appendix A: Pure Price Elasticity Restriction

- Restricted the panel strictly to border segments where  $\Delta \text{Top 50} = 0$ .
- Does the financial price surface when Caliber prestige is identical?

Table: Pure Price Model (Dependent Var: log PPSF)

|                         | (1) Single Cluster    | (2) Two-Way Cluster   |
|-------------------------|-----------------------|-----------------------|
| Tuition Gap LATE (\$1k) | -0.003<br>(0.005)     | -0.003<br>(0.013)     |
| Property Tax Rate       | -20.096***<br>(2.349) | -20.096***<br>(3.438) |
| Fixed Effects           | Seg $\times$ Year     | Seg $\times$ Year     |
| Observations            | 79,318                | 79,318                |
| $R^2$ Adj.              | 0.832                 | 0.832                 |

**Result:** A precise, perfectly-identified zero effect for raw financial cost.

## Appendix B: Marginal Tax Cliff Restriction

- Restricted panel strictly to dyads located in the lowest income tax differential quartile. Eliminates Tiebout income-tax haven sorting as a confounder.

Table: Tax Restricted Model (Dependent Var: log PPSF)

|                           | (1) Single Cluster | (2) Two-Way Cluster |
|---------------------------|--------------------|---------------------|
| Caliber Gap LATE (Top 50) | 0.085*<br>(0.048)  | 0.085<br>(0.082)    |
| Tuition Gap LATE (\$1k)   | 0.014<br>(0.009)   | 0.014<br>(0.024)    |
| Fixed Effects             | Seg $\times$ Year  | Seg $\times$ Year   |
| Observations              | 36,936             | 36,936              |
| $R^2$ Adj.                | 0.831              | 0.831               |

# Appendix C: Stability Across Timeframes

- Does the Remote Work / COVID housing boom rupture our parameters?

Table: Pre-2019 vs Post-2020 Parameters (Two-way Cluster)

|                           | (1) Pre-COVID       | (2) Post-COVID    |
|---------------------------|---------------------|-------------------|
| Caliber Gap LATE (Top 50) | 0.118***<br>(0.040) | 0.062<br>(0.041)  |
| Tuition Gap LATE (\$1k)   | -0.015<br>(0.015)   | -0.001<br>(0.010) |
| Fixed Effects             | Seg $\times$ Year   | Seg $\times$ Year |
| Observations              | 78,046              | 50,374            |
| $R^2$ Adj.                | 0.773               | 0.833             |

**Result:** Model handles secular shocks well; the Top 50 premium operates strongly across time, though it compresses slightly during national housing spikes.